FactLens: Detecting Fake News and Political Bias with Explainable ML

Mauricio Velasquez · Valentina Kloster · John Corbacho Soubal
mvela047@fiu.edu · vklos001@fiu.edu · jcorb019@fiu.edu

Abstract

FactLens is a dual-task ML system for detecting fake news and political bias in news articles. Logistic Regression (TF-IDF) and fine-tuned DistilBERT were trained on two Kaggle datasets (38,590 and 13,366 articles). DistilBERT reached 99.97% on fake news; both models achieved ~87% on political bias. A source bias experiment confirmed genuine content learning beyond formatting signals. Cross-task analysis (Pearson r = -0.049, zero shared top-20 vocabulary) showed that fake news and political bias are linguistically independent, requiring separate specialized models.

Introduction

Misinformation spreads six times faster than truth online [1], and 75% of people cannot reliably identify fake news [2]. Political bias presents a related but distinct challenge: the same facts, framed differently, produce different impressions. This paper addresses three questions: (1) Can an interpretable pipeline reliably detect fake news? (2) Can a model detect political bias from article language? (3) Do fake news and political bias share overlapping linguistic patterns? The third question is our original contribution, directly testing whether these two phenomena are linguistically connected.

Methodology

Text was preprocessed through lowercasing, stopword removal (NLTK), lemmatization, and duplicate removal. TF-IDF vectorization (50,000 features) was applied with an 80/20 stratified split. Five linguistic features were also extracted: VADER sentiment, TextBlob subjectivity, Flesch Reading Ease, exclamation count, and question count, scaled with StandardScaler. Logistic Regression (L-BFGS, C=1.0, balanced class weights) was trained on the combined matrix; its coefficients provide direct interpretability. A source bias experiment removed 30 Reuters-linked tokens to separate formatting from content signals. DistilBERT (distilbert-base-uncased, 67M parameters) was fine-tuned for 3 epochs (AdamW, FP16) on a Colab A100. Explainability used Integrated Gradients (Captum); LIME produced near-zero scores due to model confidence exceeding 99.9%, a known limitation on highly confident classifiers [7]. Cross-task analysis used Pearson and Spearman correlations on shared LR coefficient vectors, top-20 vocabulary overlap, and KS tests on linguistic features.

Dataset

D1 [3]: 38,590 articles after deduplication (45.1% Fake, 54.9% Real); Real from Reuters, Fake from political blogs. Fake articles used exclamation marks 12x more and had higher subjectivity (0.454 vs. 0.362). D2 [4]: 13,366 articles after removing Center labels (58.4% Left, 41.6% Right), with wider source diversity and residual outlet-name bias in LR coefficients. Full statistics are in Appendix A and B.

Simulation and Results

All experiments ran in Google Colab (Python 3.12). Tables 1 and 2 summarize results. The 0.68% train-test gap confirms no significant overfitting (AUC = 0.9985). The debiased LR dropped only 1.02%, confirming genuine content learning. DistilBERT's near-perfect accuracy (99.97%) reflects genuine learning alongside residual formatting patterns. On D2, both models tied at ~87%, indicating political bias is keyword-driven and does not benefit from contextual modeling.

Original LR top fake tokens were formatting artifacts ("reuters" +19.60, "said" +18.86); after debiasing, signals shifted to genuine misinformation language ("president donald" +8.51, "dont" +7.23). Integrated Gradients confirmed "reuters" as the strongest Real signal (0.8578). Cross-task: Pearson r = -0.049 (p = 0.145); zero shared tokens in any top-20 class pair; KS tests showed no consistent political lean in fake news. Fake news and political bias are orthogonal and require separate models.

Model	Accuracy	Precision	Recall	F1
LR - Original (TF-IDF)	98.52%	0.99	0.98	0.99
LR - Debiased	97.50%	0.98	0.97	0.97

LR - Enhanced (+5 feat.)	98.33%	0.98	0.98	0.98
DistilBERT - Fine-tuned	99.97%	1.00	1.00	1.00

Table 1. Model performance on D1 (Fake vs. Real, n = 7,718 test articles).

Model	Accuracy	Precision	Recall	F1
LR - L1 (C=5)	87.06%	0.87	0.87	0.87
DistilBERT - Fine-tuned	86.60%	0.87	0.86	0.87

Table 2. Model performance on D2 (Left vs. Right, n = 2,674 test articles).

Conclusion

FactLens showed that interpretable ML can reliably detect both fake news and political bias. The debiasing experiment confirmed genuine content learning beyond source formatting. Integrated Gradients proved more informative than LIME for highly confident models. The LR-DistilBERT tie on D2 suggests political bias is vocabulary-driven, making deep contextual modeling unnecessary. Most importantly, the cross-task analysis confirmed these are linguistically independent phenomena that require separate classifiers. Future work could extend to three-class bias (Left/Center/Right), apply debiasing to the transformer pipeline, and test on more source-diverse datasets.

References

[1] Vosoughi, S., Roy, D., & Aral, S. (2018). The spread of true and false news online. Science, 359(6380), 1146-1151.

[2] Pennycook, G., & Rand, D. G. (2019). Fighting misinformation on social media using crowdsourced judgments of news source quality. PNAS, 116(7), 2521-2526.

[3] Bisaillon, C. (2020). Fake and real news dataset. Kaggle. https://www.kaggle.com/datasets/clmentbisaillon/fake-and-real-news-dataset

[4] cl0ud0. (2022). News political bias classification dataset. Kaggle. https://www.kaggle.com/datasets/cl0ud0/news-political-bias-classification-dataset

[5] Sanh, V. et al. (2019). DistilBERT, a distilled version of BERT. arXiv:1910.01108.

[6] Sundararajan, M. et al. (2017). Axiomatic attribution for deep networks. ICML 2017.

[7] Ribeiro, M. T. et al. (2016). "Why should I trust you?" KDD 2016.

Appendices

Appendix A - Dataset Statistics

Dataset	Source	Total	Class split
D1 - Fake/Real	Kaggle/Bisaillon	38,590	45.1% Fake / 54.9% Real
D2 - Bias	Kaggle/cl0ud0	13,366	58.4% Left / 41.6% Right

Table A1. Dataset summary after preprocessing.

Appendix B - Linguistic Feature Means by Class

Feature	Fake	Real	Left	Right
Sentiment (VADER)	-0.075	+0.072	-0.041	-0.031

Subjectivity (TextBlob)	0.454	0.362	0.381	0.374
Readability (Flesch)	52.36	42.31	44.12	43.88
Exclamation count	0.744	0.062	0.089	0.103
Question count	1.235	0.100	0.215	0.198

Table B1. Mean linguistic feature values across all four classes.

Appendix C - Top Predictive Tokens (LR Coefficients)

Top Fake (Original)	Top Fake (Debiased)	Top Left	Top Right
reuters (+19.60)	president donald (+8.51)	nato	hegseth
said (+18.86)	dont (+7.23)	alarmed	praised
washington (+8.53)	statement (+6.84)	unpredictable	republican
wednesday (+5.10)	spokesman (+5.92)	climate	washington examiner
president donald (+4.99)	im (+5.44)	progressive	daily wire

Table C1. Top 5 predictive tokens per class from LR coefficient analysis.

Appendix D - Integrated Gradients: Top Token Attributions (DistilBERT)

Real Token	Attr. Score	Fake Token	Attr. Score
reuters	0.8578	think	0.1653
brussels	0.5111	trump	0.0939
washington	0.4088	inc	0.0839
mexico	0.2900	year	0.0756
paris	0.2030	big	0.0748

Table D1. Top 5 Integrated Gradients token attributions per class (avg. across 10 sampled articles).

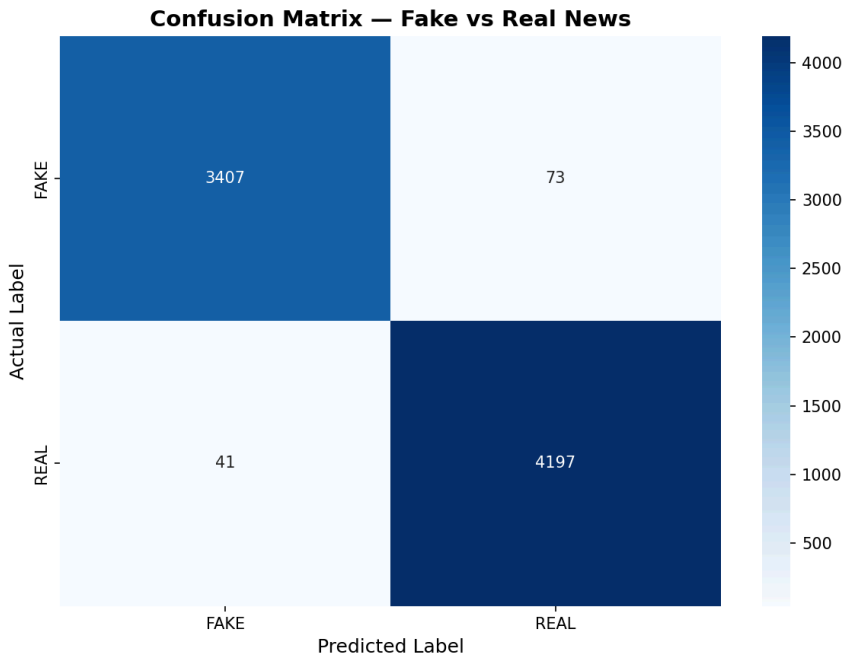
Appendix E - Cross-Task Analysis Results

Pearson correlation between LR coefficient vectors (fake news vs. political bias) across shared vocabulary: r = -0.049 (p = 0.145, not significant at alpha = 0.05). Spearman rank correlation: rho = -0.172. Vocabulary overlap: 0 shared tokens across all top-20 class pairs. KS tests on linguistic features showed no consistent political lean in fake news content across any of the five dimensions. These results confirm that fake news and political bias are orthogonal linguistic phenomena.

Appendix F - Figures

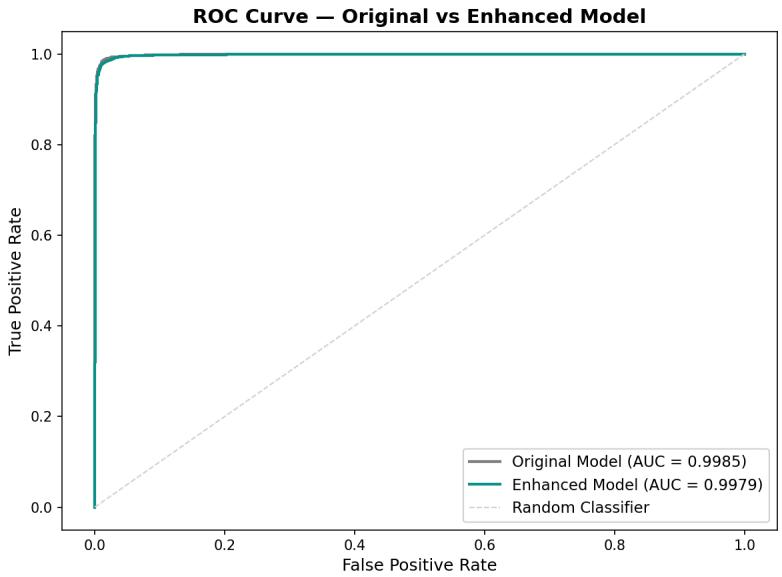
The following figures should be inserted from project outputs.

Figure F1. Confusion matrix - LR Original (D1)



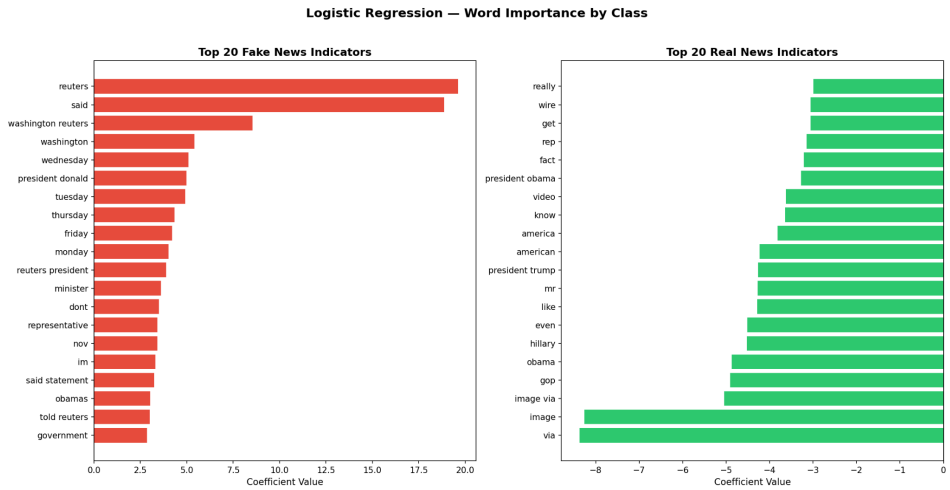
Confusion Matrix LR Original (D1) The confusion matrix shows that the Logistic Regression model correctly classified 7,604 of 7,718 test articles, with 73 Fake articles misclassified as Real and 41 Real articles misclassified as Fake. The low off-diagonal counts confirm balanced performance across both classes.

Figure F2. ROC curve - LR Original vs. Enhanced (D1)



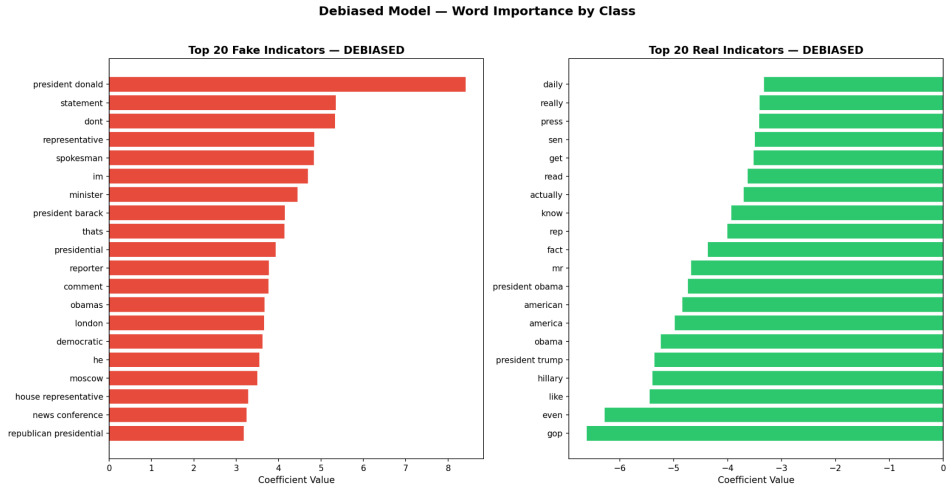
The ROC curves for the original and enhanced Logistic Regression models are nearly identical, with AUC scores of 0.9985 and 0.9979 respectively. The overlap confirms that adding five linguistic features provided no meaningful improvement in discriminative power over TF-IDF alone.

Figure F3. Top 20 Fake/Real word importance bar charts - LR coefficients original



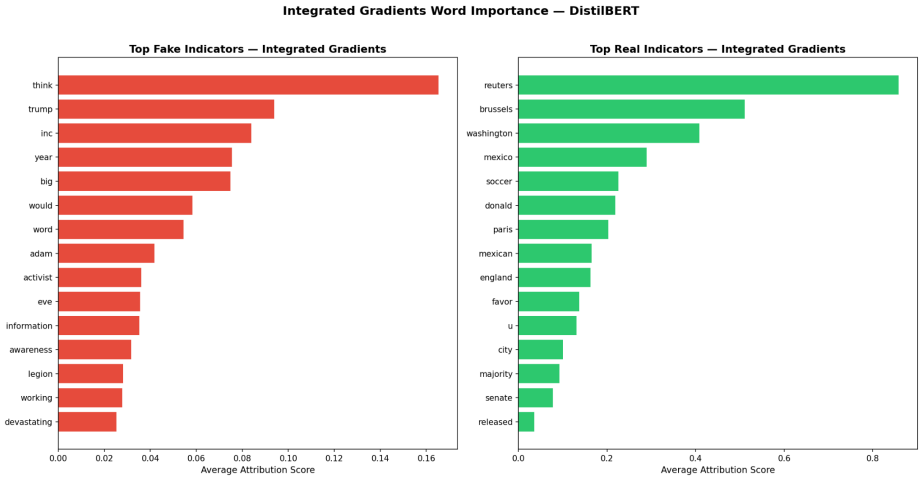
Horizontal bar charts showing the top 20 positive and negative LR coefficients from the original model. The dominant Fake indicators — "reuters", "said", "washington" — are Reuters wire service formatting tokens, revealing source bias rather than genuine misinformation signals.

Figure F4. Top 20 Fake/Real word importance bar charts - LR coefficients debiased



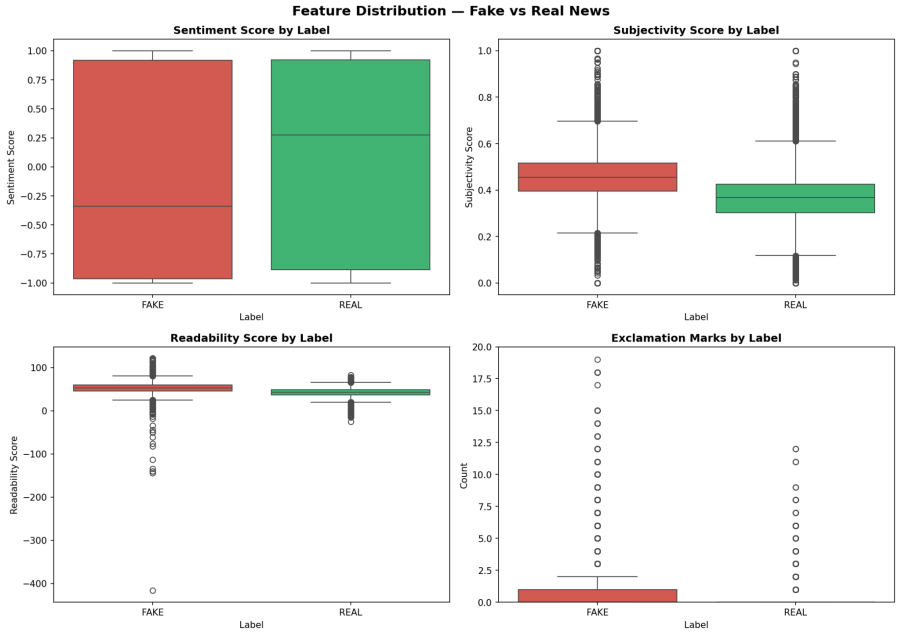
After removing 30 source-formatting tokens and retraining, the top Fake indicators shifted to substantive misinformation signals including "president donald", "dont", "statement", and "spokesman", reflecting the informal tone and political targeting characteristic of fake news content.

Figure F5. Integrated Gradients combined chart - DistilBERT (Real vs. Fake)



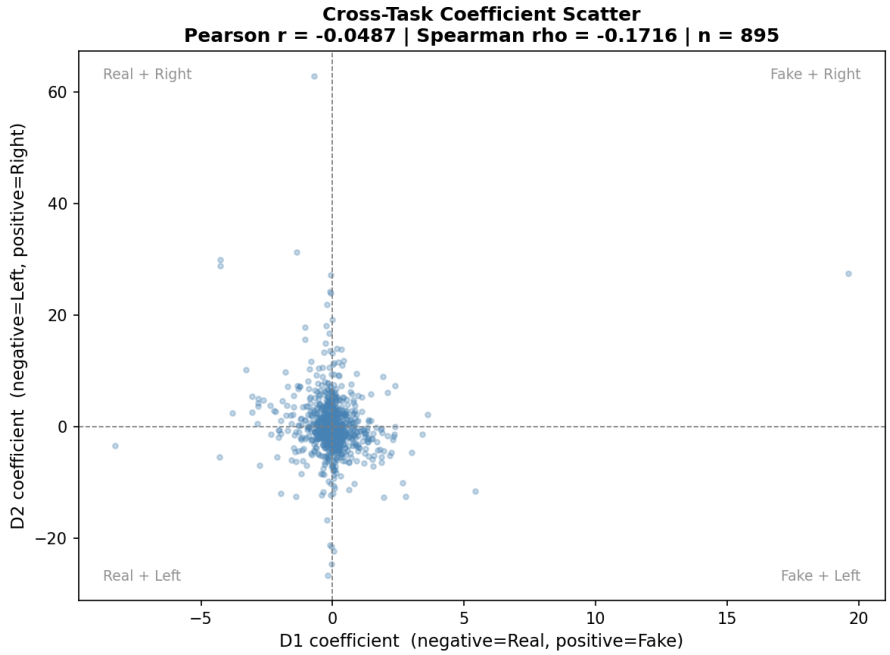
Attribution scores from Integrated Gradients averaged across 10 sampled articles show a stark asymmetry: Real articles are dominated by a single strong signal ("reuters", 0.8578) while Fake article attributions are distributed across multiple weaker content-level tokens. This confirms source bias at the transformer level and validates the debiasing experiment findings.

Figure F6. Linguistic feature distributions - box plots across Fake/Real/Left/Right



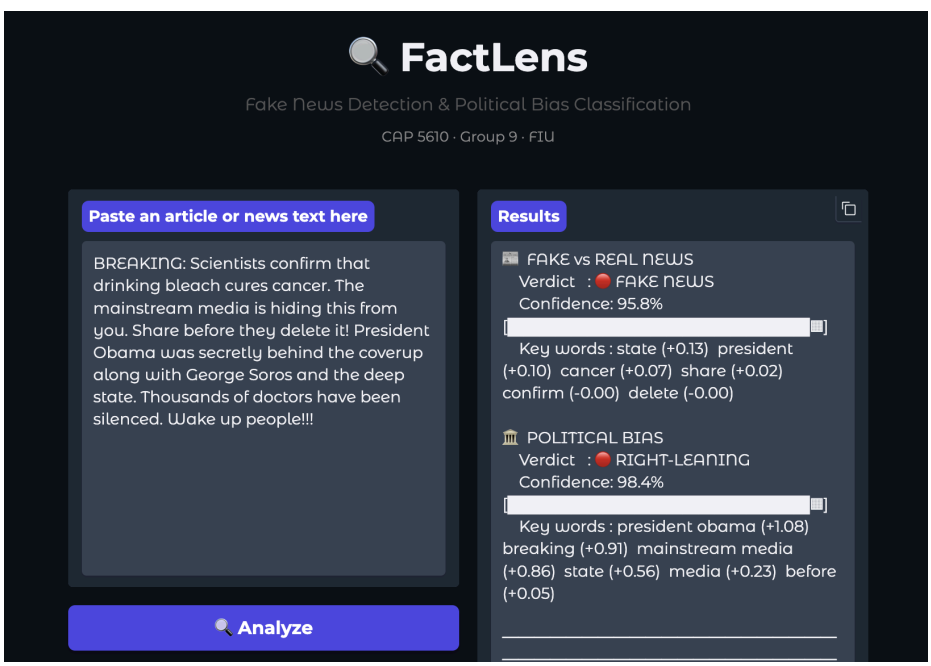
Box plots comparing five linguistic features across Fake and Real articles. Fake articles show consistently higher subjectivity (0.454 vs. 0.362), higher readability scores (52.36 vs. 42.31), and dramatically higher exclamation and question mark counts, confirming that misinformation is written in a more emotional, accessible, and rhetorically aggressive register than real journalism.

Figure F7. Cross-task coefficient scatter plot (Fake score vs. Bias score per word)



Each point represents a word appearing in both model vocabularies, plotted by its fake news coefficient (x-axis) against its political bias coefficient (y-axis). The near-zero correlation (Pearson r = −0.049) and lack of any visible trend confirm that the words driving fake news predictions are entirely unrelated to the words driving political bias predictions.

Figure F8. Gradio demo screenshot



The interactive FactLens demo classifies a real article in real time, showing simultaneous output for both tasks: Fake or Real verdict with confidence score and top contributing words, and Left or Right verdict with confidence score and top contributing words.

Figure F9. Linguistic feature distributions - Left vs. Right (box plots)

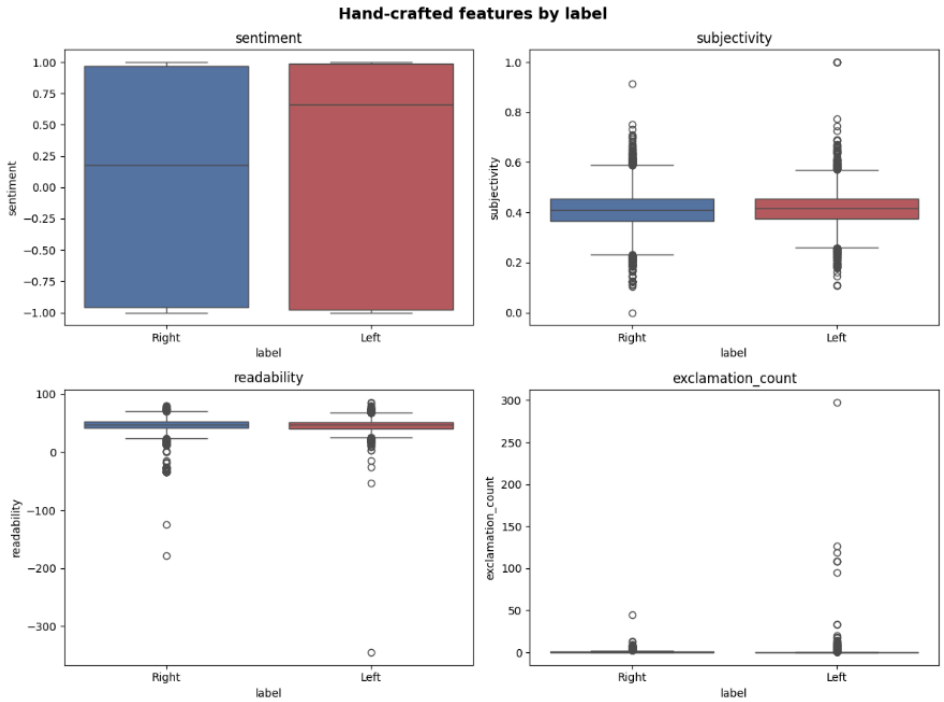
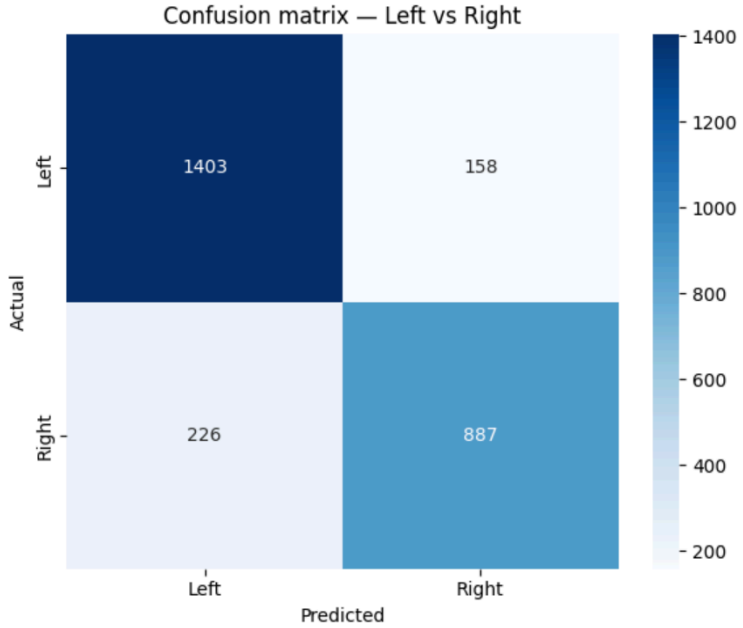


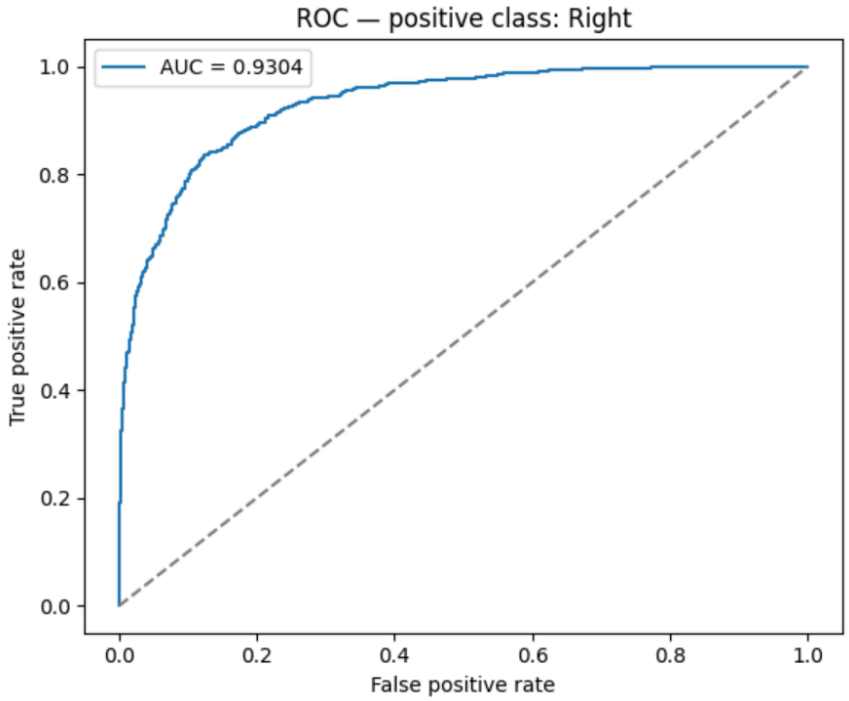
Figure F9 visualizes the distribution of four hand-crafted features (sentiment, subjectivity, readability, and exclamation count) across Left and Right articles. Consistent with Table B1, all features show substantial overlap between classes. Sentiment exhibits wide variance in both groups, with only a slight shift toward higher values in Left articles. Subjectivity distributions are nearly identical, indicating comparable levels of opinionated language. Readability scores are tightly clustered with minimal separation, suggesting similar writing complexity. Exclamation counts remain near zero for most samples, with a few extreme outliers more prominent in Right articles. Overall, the figure reinforces that individual linguistic features provide weak separation for political bias classification.

Figure F10. Confusion matrix - Logistic Regression (D2)



The confusion matrix shows 1,403 correctly classified Left articles and 887 correctly classified Right articles, with 158 Left articles misclassified as Right and 226 Right articles misclassified as Left. This asymmetry reflects the dataset imbalance (58.4% Left vs. 41.6% Right) and indicates a mild bias toward predicting the majority class. Despite this, the model maintains strong overall performance and captures the dominant patterns in both classes.

Figure F11. ROC curve - Logistic Regression (D2, positive class: Right)



The ROC curve demonstrates strong discriminative ability, with an AUC of 0.9304. The steep rise near the origin indicates that the model achieves high true positive rates with relatively low false positive rates. This confirms that, although raw feature distributions overlap, the combined feature space contains sufficient signal for effective classification.

Classification Performance Summary (D2)

The Logistic Regression model achieves an overall accuracy of 85.98%, consistent with the results reported in Table 2. Class-wise metrics show:

- Left: Precision = 0.85, Recall = 0.92, F1 = 0.89
- Right: Precision = 0.88, Recall = 0.77, F1 = 0.82

The model attains higher recall for the Left class, while Right predictions are more precise but less complete. This tradeoff aligns with the confusion matrix and reflects a conservative tendency when predicting the minority class. Macro and weighted averages (~0.85-0.86) confirm balanced performance despite class imbalance.

Appendix G - Gradio Demo

An interactive demo was built with Gradio and deployed via Google Colab (share=True). It loads two LR models and two TF-IDF vectorizers. Users paste any article text and receive: (1) Fake or Real verdict with confidence score and top contributing words; (2) Left or Right verdict with confidence score and top contributing words. Contributing words are computed at inference time by multiplying each word's

TF-IDF score by its model coefficient, ranked by absolute contribution. The demo was used live during the final presentation to classify four real-world articles in real time.